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Course/Section:CPE 212-31S1	**Date Submitted:12/11/2025**
Instructor: Engr. Robin Valenzuela	**Semester and SY: 2025-2026**

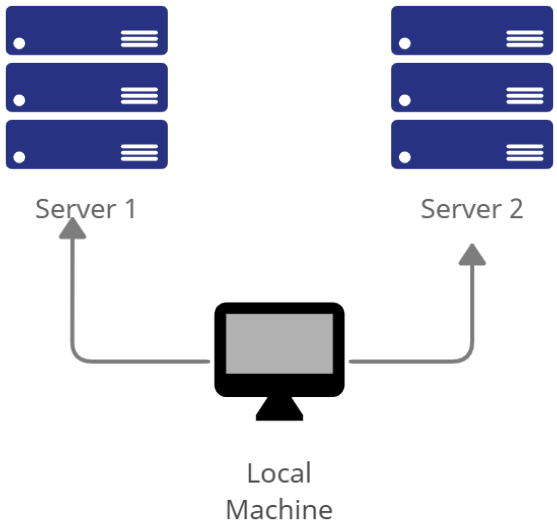
Activity 1: Configure Network using Virtual Machines

1. **Objectives:**
   1.1. Create and configure Virtual Machines in Microsoft Azure or VirtualBox
   1.2. Set-up a Virtual Network and Test Connectivity of VMs

2. **Discussion:**

Network Topology:

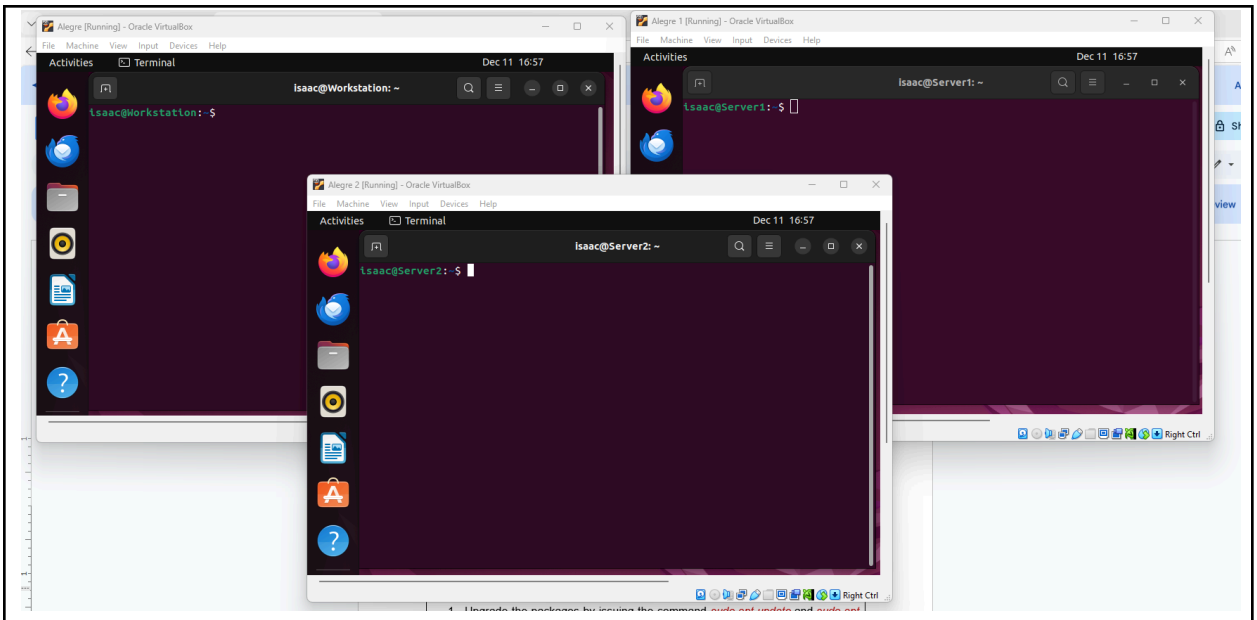
Assume that you have created the following network topology in Virtual Machines, *provide screenshots for each task*. (Note: *it is assumed that you have the prior knowledge of cloning and creating snapshots in a virtual machine*).



Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

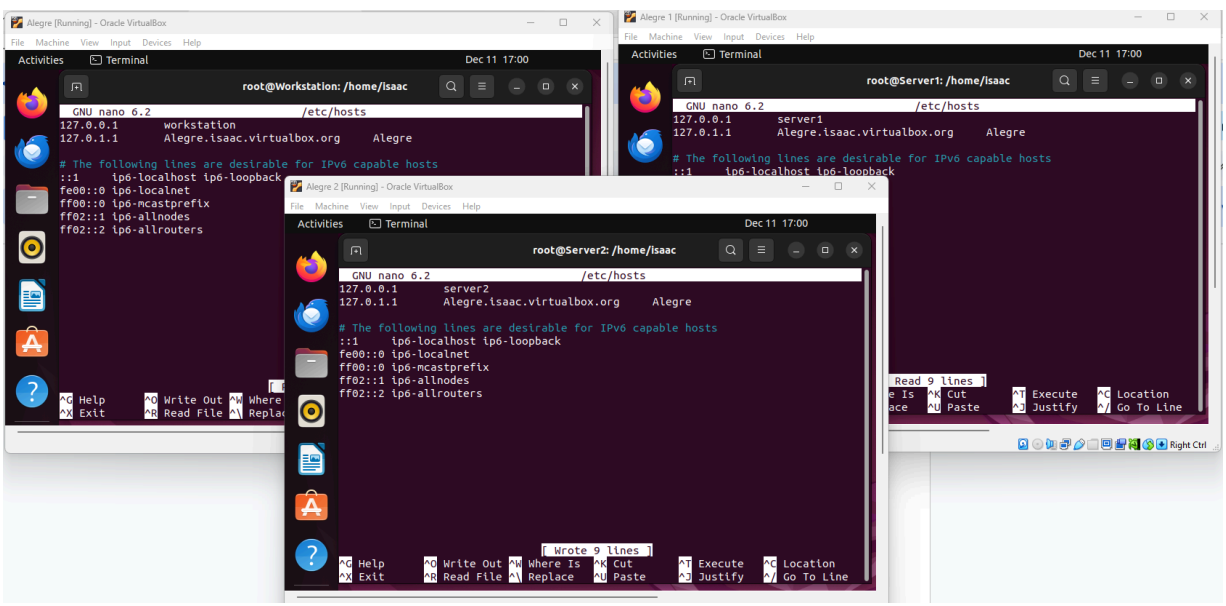
1. Change the hostname using the command *sudo nano /etc/hostname*
   1.1 Use server1 for Server 1
   1.2 Use server2 for Server 2
   1.3 Use workstation for the Local Machine

Screenshot:



2. Edit the hosts using the command *sudo nano /etc/hosts*. Edit the second line.
   2.1 Type 127.0.0.1 server 1 for Server 1
   2.2 Type 127.0.0.1 server 2 for Server 2
   2.3 Type 127.0.0.1 workstation for the Local Machine

Screenshot:

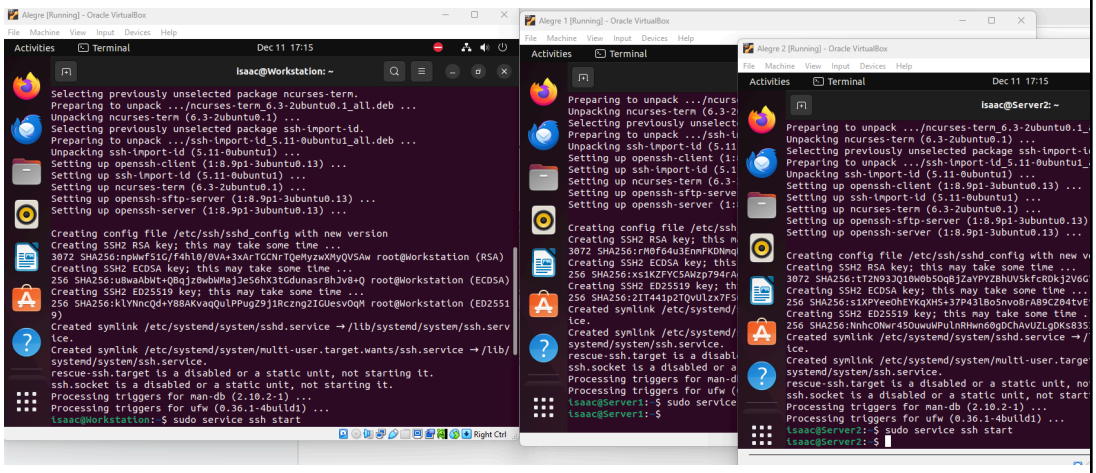


Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

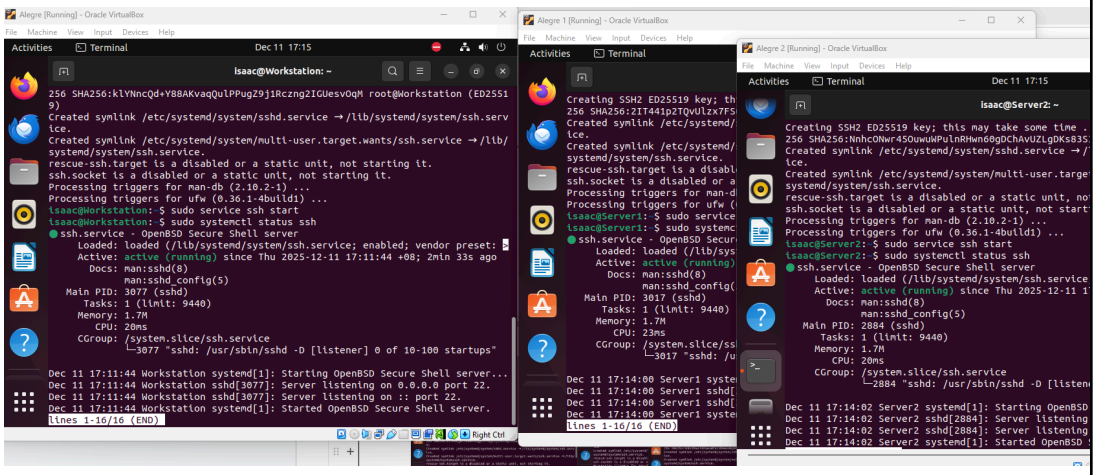
1. Upgrade the packages by issuing the command *sudo apt update* and *sudo apt upgrade* respectively.
2. Install the SSH server using the command *sudo apt install openssh-server*.

3. Verify if the SSH service has started by issuing the following commands:

3.1 *sudo service ssh start*

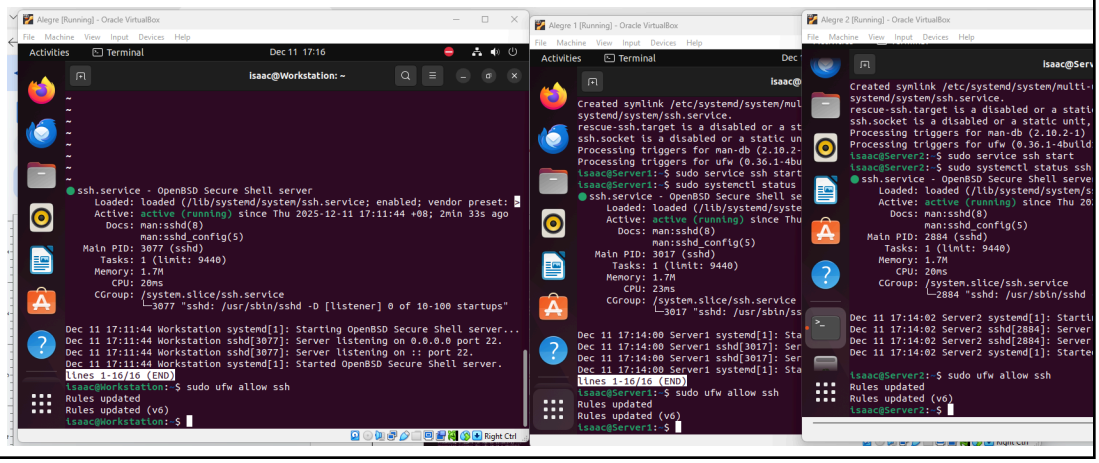


3.2 *sudo systemctl status ssh*

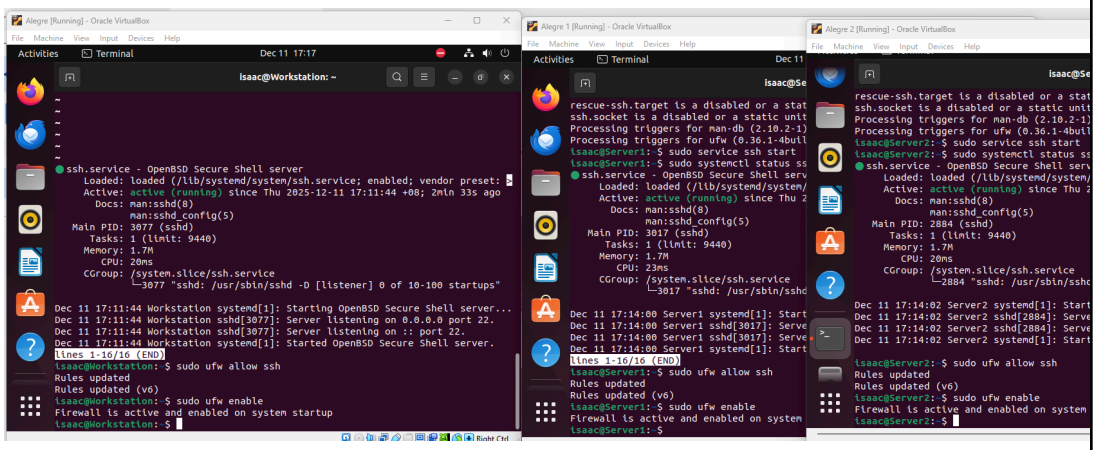


4. Configure the firewall to all port 22 by issuing the following commands:

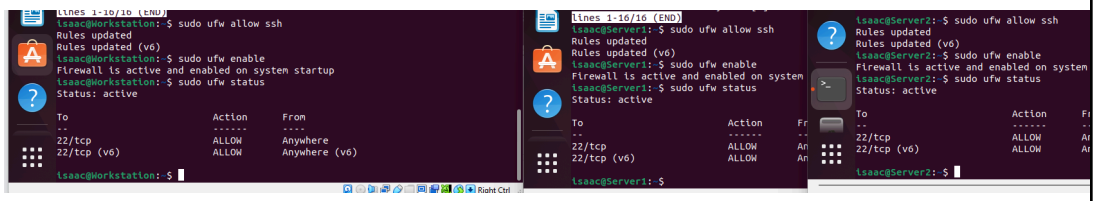
4.1 *sudo ufw allow ssh*



4.2 *sudo ufw enable*

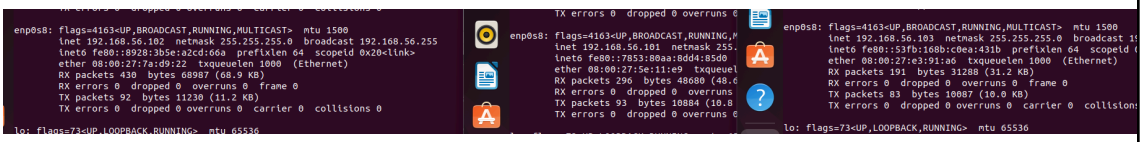


4.3 *sudo ufw status*



Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192

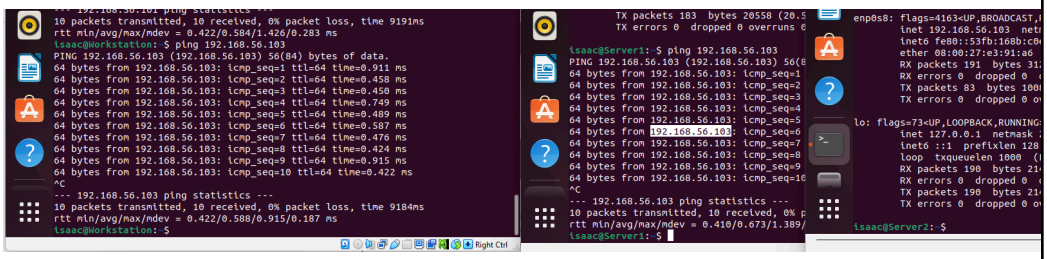


.168.56.XX.

1.1 Server 1 IP address: 192.168.56.**102**
1.2 Server 2 IP address: 192.168.56.**101**
1.3 Server 3 IP address: 192.168.56.**103**

2. Make sure that they can ping each other.
   2.1 Connectivity test for Local Machine 1 to Server 1: ☐ Successful ☐ Not Successful
   2.2 Connectivity test for Local Machine 1 to Server 2: ☐ Successful ☐ Not Successful

2.3 Connectivity test for Server 1 to Server 2: ☐ Successful ☐ Not Successful



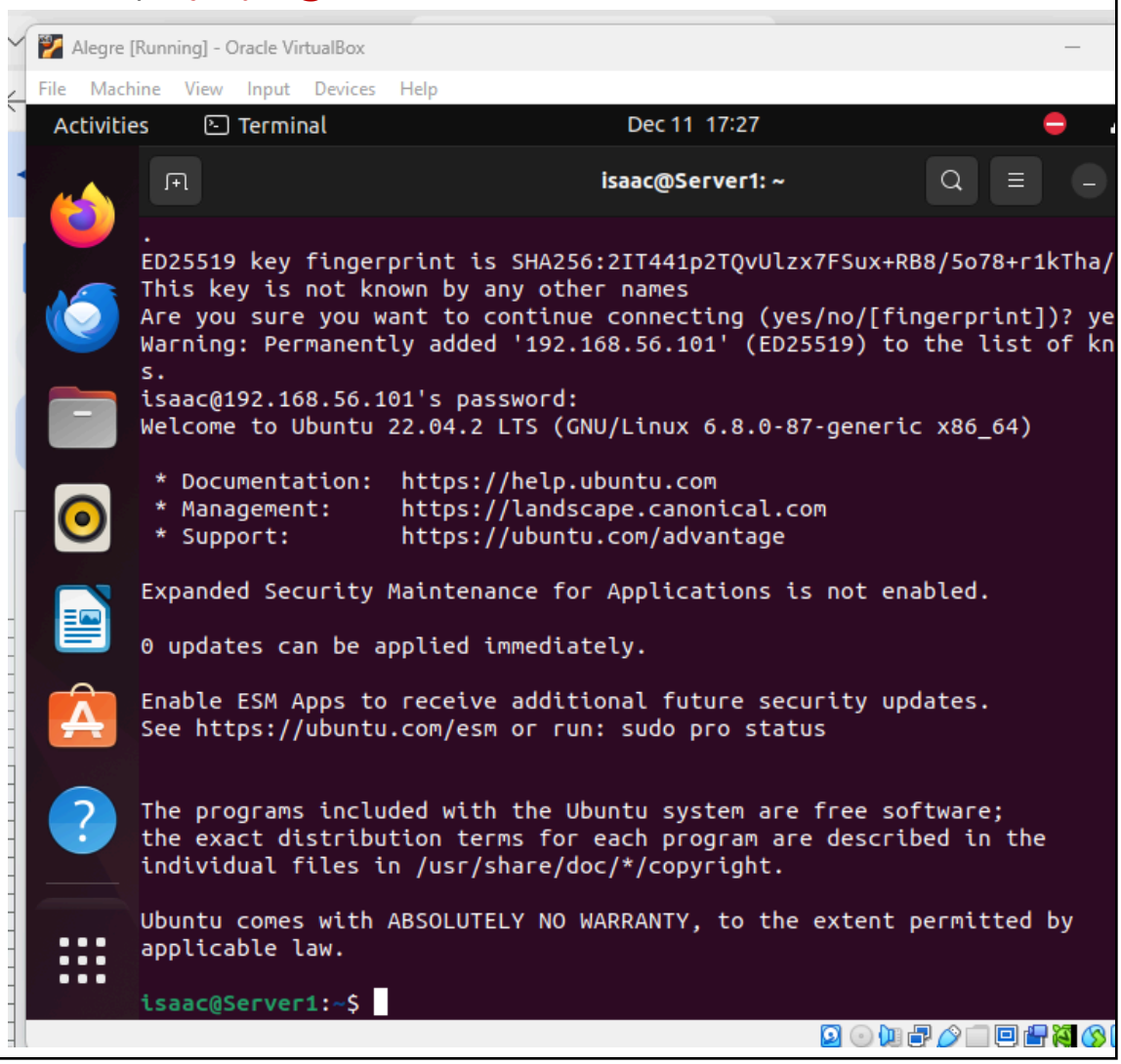
Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, *ssh jvtaylar@192.168.56.120*

1.2 Enter the password for server 1 when prompted

1.3 Verify that you are in server 1. The user should be in this format user@server1. For example, *jvtaylar@server1*



2. Logout of Server 1 by issuing the command *control + D.*

```
isaac@Server1:~$
logout
Connection to 192.168.56.101 closed.
isaac@Workstation:~$
```

3. Do the same for Server 2.

```
isaac@Server2: ~
The authenticity of host '192.168.56.101 (192.168.56.101)'
.
ED25519 key fingerprint is SHA256:2IT441p2TQvUlzx7FSux+RB8/
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[finge
Warning: Permanently added '192.168.56.101' (ED25519) to th
s.
isaac@192.168.56.101's password:
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 6.8.0-87-generic x

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

Expanded Security Maintenance for Applications is not enabl

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updat
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Dec 11 17:27:00 2025 from 192.168.56.102
isaac@Server1:~$
logout
Connection to 192.168.56.101 closed.
isaac@Server2:~$
```

4. Edit the hosts of the Local Machine by issuing the command *sudo nano /etc/hosts.* Below all texts type the following:

4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)

4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

4.3 Save the file and exit.

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do *ssh jvtaylar@server1*. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Server1:

```
isaac@Workstation:~$ sudo nano /etc/hosts
isaac@Workstation:~$ ssh isaac@server1
The authenticity of host 'server1 (192.168.56.101)' can't be established.
ED25519 key fingerprint is SHA256:2IT441p2TQvUlzx7FSux+RB8/5o78+r1kTha/2Go
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server1' (ED25519) to the list of known hosts.
isaac@server1's password:
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

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New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Dec 11 17:29:06 2025 from 192.168.56.103
isaac@Server1:~$
```

Server2:

```
isaac@Workstation:~$ ssh isaac@server2
The authenticity of host 'server2 (192.168.56.103)' can't be established.
ED25519 key fingerprint is SHA256:NnhcONwr45OuwuWPulnRHwn60gDChAvUZLgDKs83S
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server2' (ED25519) to the list of known hosts.
isaac@server2's password:
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

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The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

isaac@Server2:~$
```

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
   I was able to do it by declaring the hostname and ip address of server1 and server2 inside the hosts of workstation.
2. How secured is SSH?

   It is secured by the encryption feature it has and also the authentication on remote.